

CORE PURE / COMPLEX NUMBERS

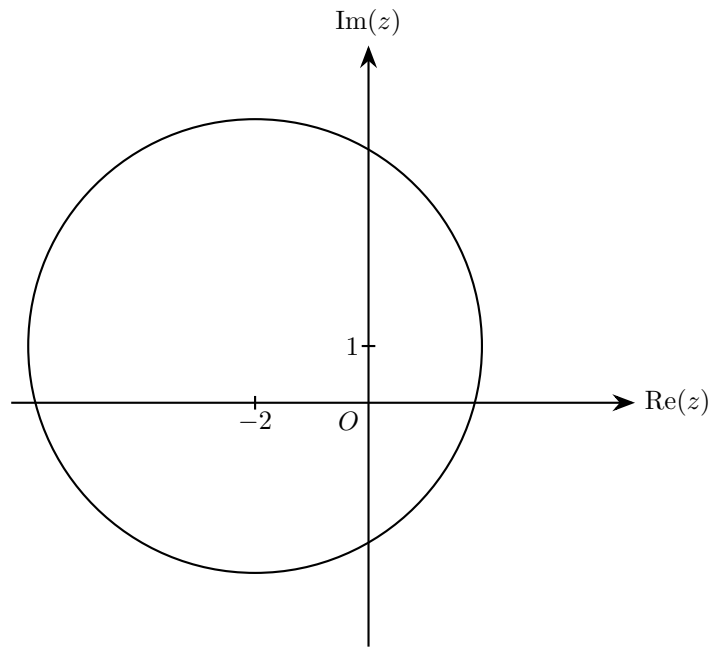
Loci and Regions in the Argand Diagram

Questions by topic

10 questions · 88 marks

Questions

Question 1	6 marks	2
Question 2	9 marks	3
Question 3	10 marks	4
Question 4	12 marks	5
Question 5	12 marks	6
Question 6	12 marks	7
Question 7	7 marks	8
Question 8	7 marks	9
Question 9	7 marks	10
Question 10	6 marks	11



1. The Argand diagram shows a circle of radius 4. The centre of the circle is the point which represents the complex number $-2 + i$.

- (a) Use set notation to define the locus of complex numbers, z , represented by points which lie on the circle. [2]

The locus L is defined by

$$L = \left\{ z \in \mathbb{C} \mid |z - (2 + i)| = |z + 1 - 3i| \right\}$$

- (b) On a copy of the Argand diagram, sketch and label the locus L . [2]

You are given that the locus

$$\left\{ z \in \mathbb{C} \mid \arg(z + 1 - 2i) = \frac{2\pi}{3}, \operatorname{Im}(z) = 5 \right\}$$

contains only one number.

- (c) Find this number. [2]

2. In an Argand diagram, the point P representing the complex number w lies on the locus defined by

$$\left\{ z \in \mathbb{C} \mid \arg(z - 2 + i) = \frac{\pi}{6} \right\}$$

You are given that $\text{Im}(w) = 3$.

(a) Find w .

[2]

The point P also lies on the locus defined by $\left\{ z \in \mathbb{C} \mid |z - 2 - 11i| = k \right\}$, where k is a constant.

(b) Find the complex number represented by the other point of intersection of the loci defined by

$$\left\{ z \in \mathbb{C} \mid |z - 2 - 11i| = k \right\} \quad \text{and} \quad \left\{ z \in \mathbb{C} \mid \arg(z - 2 + i) = \frac{\pi}{6} \right\}$$

[7]

3. On an Argand diagram, the locus of points satisfying $|z - 4 + 3i| = 8$ is a circle.

(a) Write down

(i) the coordinates of the centre of this circle [1]

(ii) the radius of this circle [1]

(b) On an Argand diagram, shade the set of points defined by

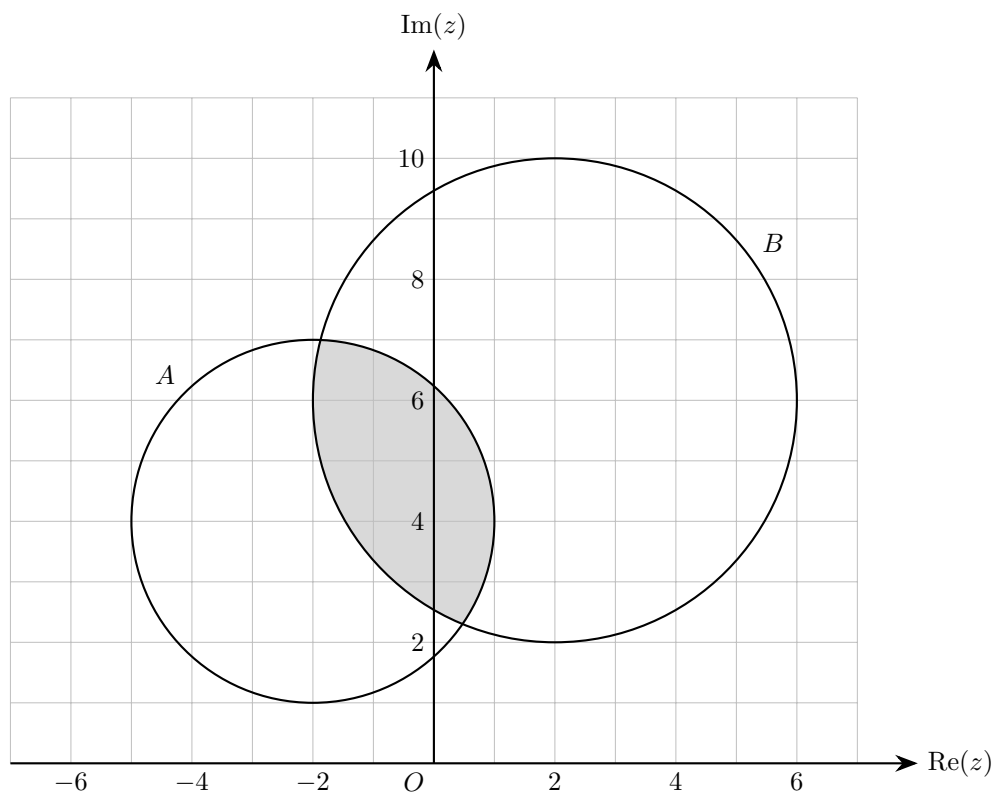
$$|z - 4 + 3i| \leq 8 \quad [1]$$

(c) For the set of points defined in part (b), determine the maximum value of $|z|$. [3]

The set of points A is defined by

$$A = \left\{ z \in \mathbb{C} \mid -\frac{\pi}{4} \leq \arg(z - 3 - 6i) \leq \frac{3\pi}{4} \right\} \cap \left\{ z \in \mathbb{C} \mid |z - 4 + 3i| \leq 8 \right\}$$

(d) Determine the area of the region defined by A , giving your answer to 3 significant figures. [4]



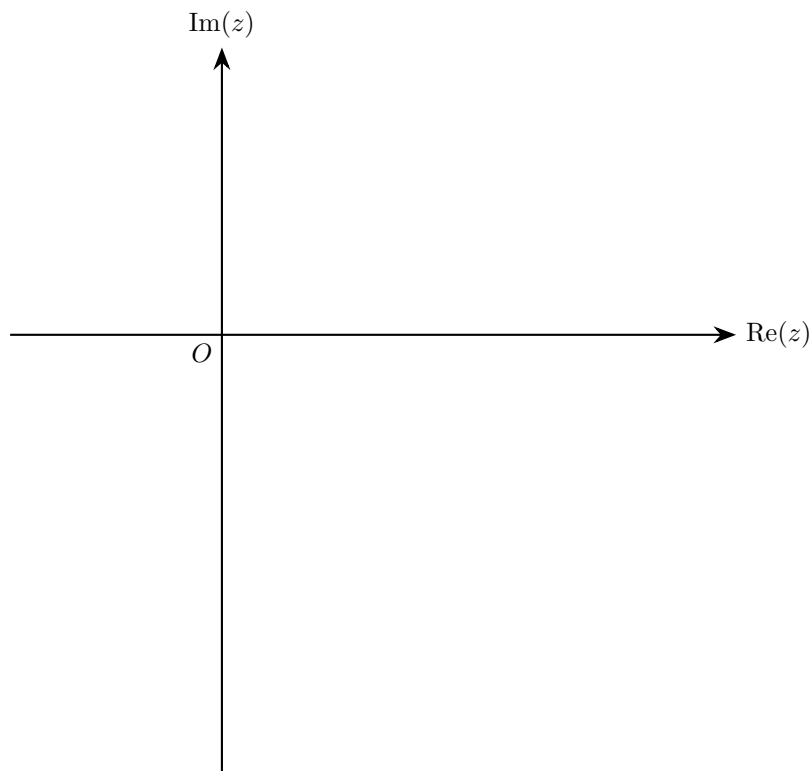
4. (a) The diagram above shows an Argand diagram.

- (i) Write down the equation of the locus represented by the circumference of circle A . [3]
- (ii) Write down the two inequalities that define the shaded region, not including either circumference. [3]

(b) (i) Draw an Argand diagram to show the region where

$$-\frac{\pi}{6} < \arg(z + 2 - i) < \frac{\pi}{4} \quad [3]$$

- (ii) Determine whether the point $1 + 5i$ lies within this region. [3]



5. (a) Find the modulus of the complex number $2 + 2\sqrt{3}i$, giving your answer as an integer. [2]
- (b) The locus of points, L , satisfies the equation $|z - (2 - 2\sqrt{3}i)| = 2$. [3]
- (i) Sketch the locus L on the Argand diagram above. [3]
- (ii) The complex number w lies on L so that $-\pi < \arg w \leq \pi$. Find the greatest possible value of $\arg w$, giving your answer in terms of π . [2]
- (c) Solve the equation $z^4 = 2 + 2\sqrt{3}i$, giving your answers in the form $re^{i\theta}$, where $r > 0$ and $-\pi < \theta \leq \pi$. [5]

6. The point P represents a complex number z on an Argand diagram such that

$$|z - (1 + i)| = \frac{1}{2}|z - (4 + i)|$$

- (a) Show that, as z varies, the locus of P is a circle, and give the coordinates of the centre and the radius of the circle. [5]

The point Q represents a complex number z on an Argand diagram such that

$$|z - (2 + 3i)| = |z + 2 - i|$$

- (b) Sketch, on the same Argand diagram, the locus of P and the locus of Q as z varies. [5]

- (c) On your diagram shade the region which satisfies

$$|z - (1 + i)| \leq \frac{1}{2}|z - (4 + i)| \quad \text{and} \quad |z - (2 + 3i)| \geq |z + 2 - i| \quad [2]$$

7. On an Argand diagram, the point P representing the complex number z lies on the locus

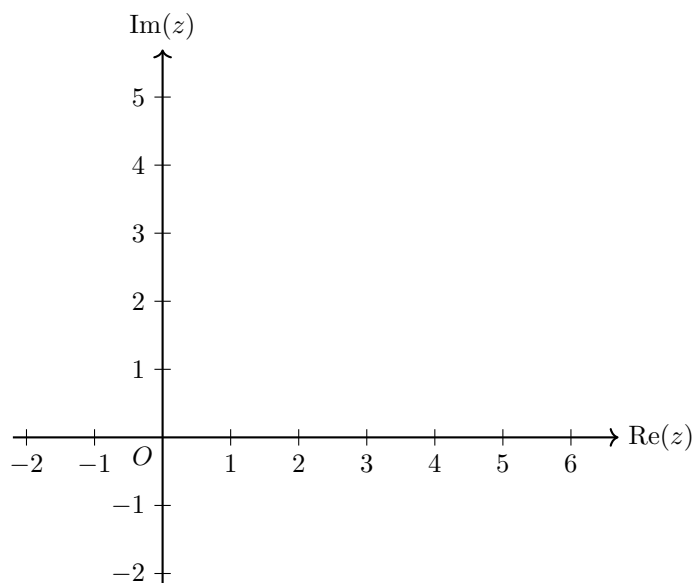
$$|z - (2 - i)| = 3$$

The point Q representing the complex number w lies on the locus

$$|w + 1 - 11i| = 3|w + 1 - 3i|$$

Find the exact coordinates of the points common to these two loci.

[7]



8. (a) Sketch, on the Argand diagram above, the locus of points satisfying the equation

$$|z - (3 + i)| = 1 \quad [1]$$

- (b) There is a unique complex number w that satisfies both

$$|w - (3 + i)| = 1 \quad \text{and} \quad \arg(w) = \alpha$$

where α is a constant such that $0 < \alpha < \pi$.

- (i) Find the value of α . [2]
(ii) Find w and express it in the form $r(\cos \theta + i \sin \theta)$. Give each of r and θ to two significant figures. [4]

9. The locus of points $z = x + iy$ that satisfy

$$\arg \left(\frac{z - (2 + i)}{z - (6 + 5i)} \right) = \frac{\pi}{4}$$

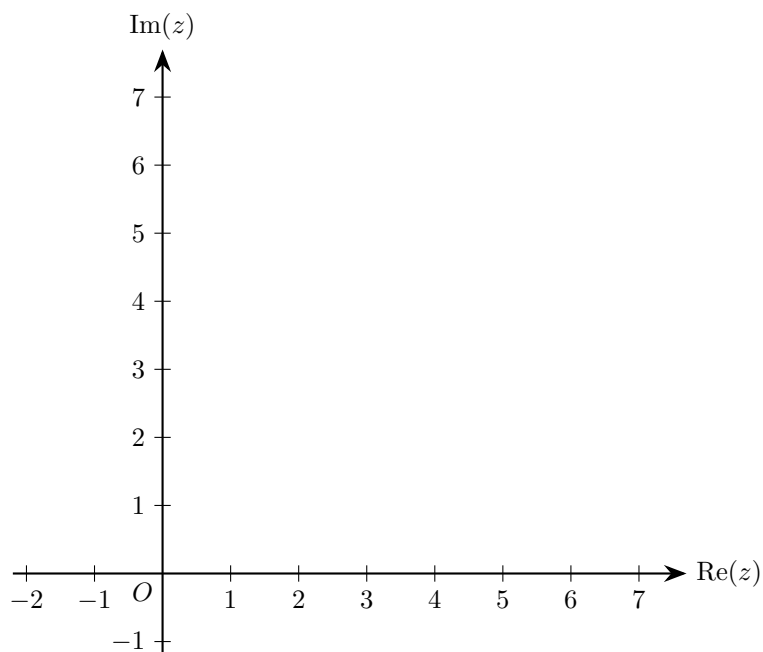
is an arc of a circle C .

(a) On an Argand diagram sketch the locus of z . [2]

(b) Explain why the centre of C lies on the line $y = 7 - x$. [1]

(c) Determine the radius of C . [2]

(d) Determine the coordinates of the centre of C . [2]



10. (a) Sketch, on the Argand diagram above, the locus of points satisfying the equation

$$|z - (2 + 4i)| = 3 \quad [2]$$

- (b) Sketch, also on the Argand diagram, the locus of points satisfying the equation

$$\arg(z - 1) = \frac{\pi}{4} \quad [1]$$

- (c) For the complex number w , find the maximum value of $|w|$ such that

$$|w - (2 + 4i)| \leq 3 \quad \text{and} \quad 0 \leq \arg(w - 1) \leq \frac{\pi}{4} \quad [3]$$